

SMART INDIA HACKATHON 2026

Problem Statement ID	: SIH26184
Problem Statement Title	: Predictive Analysis Framework for Proactive Cyber Crime Intervention
Theme	: Blockchain and Cyber Security
PS Category	: Software
Team Name (Registered on portal)	: LOSERS

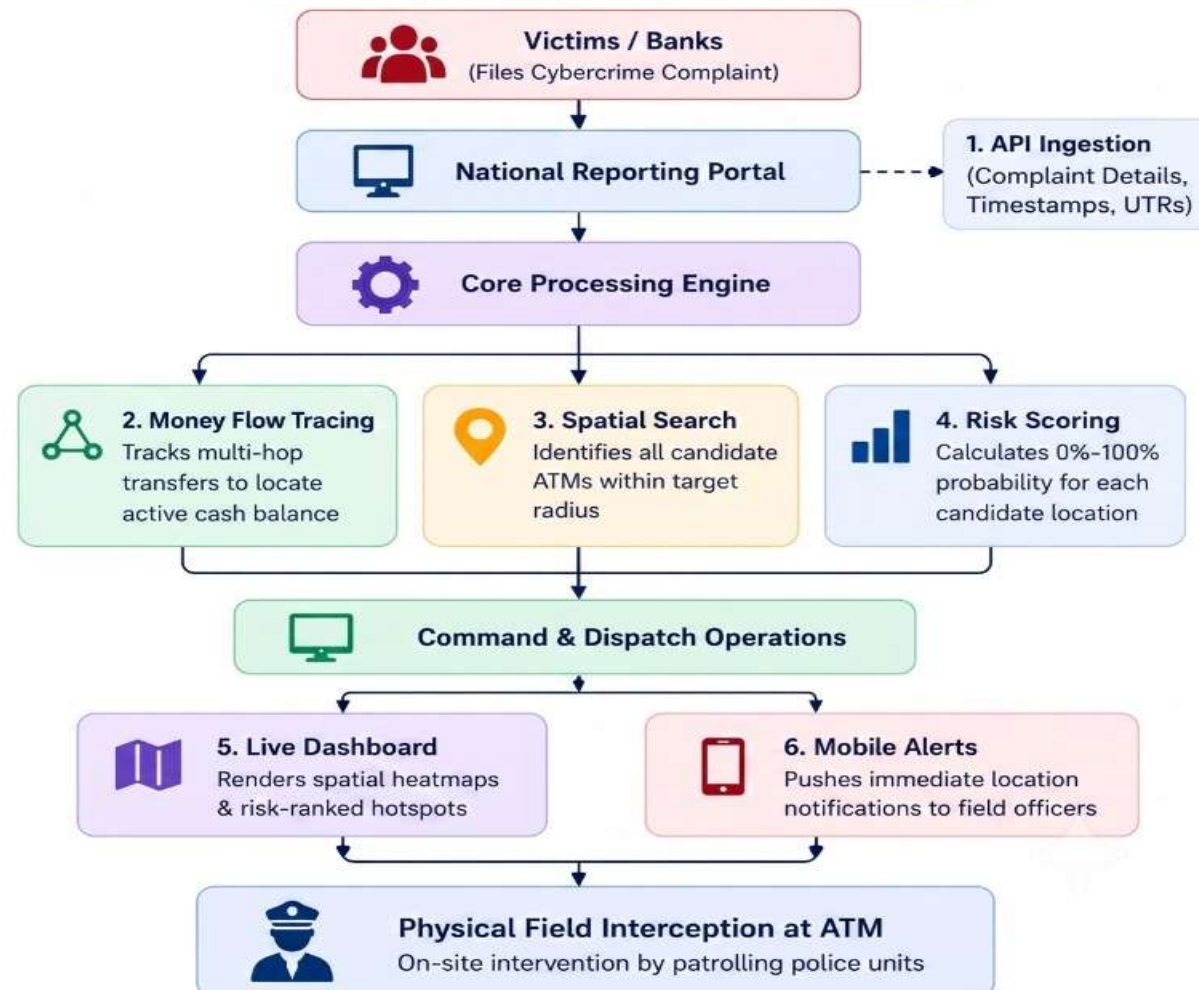
PROPOSED SOLUTION

A prediction intelligence platform for proactive cybercrime cashout intervention

Anticipating fraud cashouts before they happen

- 1 Data Ingestion & Integration Layer**
 - Ingests financial complaint feeds from portals like national cybercrime helplines via REST APIs.
- 2 Multi-Hop Graph Tracing Engine**
 - Construct account-to-account transaction graphs.
- 3 Spatial-Temporal Risk Engine:**
 - Evaluates candidate cash-out points within a target radius, assigning risk scores based on proximity, time elapsed, historical fraud frequency, and location characteristics.
- 4 Command & Control Dashboard**
 - Displays predictive spatial heatmaps and risk-ranked ATM pins on interactive maps for control-room operators.
- 5 Automated Field Dispatch**
 - Pushes real-time map updates via WebSockets and sends targeted SMS/push alerts to patrolling police units near high-risk ATMs to enable on-site interception.

WORK FLOW OF THE PROPOSED SOLUTION



Proactive Cybercrime Cashout Intervention
Physical interception at the predicted high-risk ATM
by nearest patrol units.

FEASIBILITY, VIABILITY & RISK STRATEGY



FEASIBILITY



Easy to Build & Deploy

Lightweight setup (FastAPI + PostgreSQL).



Seamless Integration

Data via REST APIs; no changes to existing systems.



Fast Processing

Graph & spatial analysis in < 2 sec, fits 15–30 min window.



Scalable Design

Handles more complaints with standard database replicas.



Existing Helpline / Bank Systems



REST Webhooks



TraceGuard Platform



Actionable Intelligence



VIABILITY



Saves More Money

On-site recovery before withdrawal prevents large fraud losses.



Smarter Patrols

Focuses police on high-risk ATMs, not across thousands of locations.



Breaks Syndicates

Catches mule handlers with multiple cards/phones and stops networks early.



Target Patrols to High-Risk ATM Hotspots



On-site Interception & Cash Recovery



Syndicate Disruption & Reduced Crime Rate



POTENTIAL CHALLENGES, RISKS & MITIGATION STRATEGIES

1

Reporting Delays



Risk:

Victims report late, missing the 15–30 min window.

Mitigation:

Track repeated mule accounts and predict next cash-out attempts.

2

False Positives & Resource Strain



Risk:

Low-confidence alerts waste police resources.

Mitigation:

Trigger field alerts only if risk > 80%; for 50–79%, apply bank holds instead.

3

Mule Mobility & Evasion



Risk:

Mules move across ATMs or use CSPs and micro-ATMs.

Mitigation:

Use spatial clustering (DBSCAN) to alert on 2–5 km hotspot clusters, not single ATMs.

4

Data Privacy & PII Compliance



Risk:

Live complaint data may breach privacy laws.

Mitigation:

Anonymize all PII using hashing (Hash Acc) at ingestion before analytics.

IMPACT & BENEFITS

KEY IMPACTS (MEASURABLE TRANSFORMATION)



Closes the Response Time-Gap:
Shifts law enforcement from multi-day reactive paper trail to a 15–30 minute proactive window before cash leaves the ecosystem.



Direct Syndicate Neutralization:
Enables physical arrests of active “mule handlers” at ATMs, securing debit cards, mobile devices, and stolen cash.



Dramatically Reduces Capital Loss:
Stops illicit cash drain at the final withdrawal stage, preserving citizens’ funds before they enter untraceable cash networks.



Data-Driven Patrol Optimization:
Replaces blind, unguided city-wide patrols with targeted deployments to AI-identified high-probability ATM clusters.

CORE BENEFITS (STAKEHOLDER VALUE)

STAKEHOLDER		KEY VALUE DELIVERED
	Law Enforcement	Real-time command-center map dashboard with automated SMS route alerts sent directly to patrolling field officers.
	Citizens & Victims	Exponentially higher probability of recovering stolen funds before money-mules complete physical withdrawals.
	Banking Ecosystem	Automated intelligence sharing that flags interconnected money-mule account networks across multiple institutions.
	Government Agencies	Fully compliant, privacy-preserving analytical layer (built with PII tokenization) that plugs directly into existing 1930 Helpline infrastructure.

BEFORE (REACTIVE APPROACH)



VS

AFTER (PROACTIVE APPROACH WITH TRACEGUARD)



Outcome: Faster Action • Higher Recovery • Stronger Enforcement • Safer Communities

RESOURCES

1. PRIMARY KNOWLEDGE & DOMAIN SOURCES



I4C & MHA Guidelines

Studied 1930 Helpline protocols and CFCFRMS operational guidelines.



National Cyber Crime Reporting Portal

Analyzed standard complaint fields (UTRs, timestamps, accounts) to design API data models.



Open Financial & Spatial Standards

Used ISO 20022 (transfer chains) and geospatial standards (Geohash / PostGIS ST_DWithin).



Indian Cybercrime Infrastructure

Aligned solution with real Indian cybercrime reporting and coordination systems.

2. DEVELOPMENT RESOURCES (TECHNICAL STACK)



PostgreSQL + PostGIS

Stores complaints, transactions, ATM & location data. PostGIS used for spatial queries to find nearby candidate ATMs.



NetworkX

Builds transaction graphs and traces multi-hop money flows to identify mule networks.



XGBoost

ML model for spatial-temporal risk scoring of candidate ATMs.



FastAPI

Backend framework for REST APIs, webhooks and real-time data processing.



React.js + Mapbox GL JS

Interactive dashboard with real-time maps and risk heatmaps for command center.



Twilio API

Sends automated SMS alerts to nearby field officers.



KEY TAKEAWAY

We combine proven Indian cybercrime infrastructure knowledge with robust, open-source technologies to deliver real-time, actionable intelligence.